

Sample Project Documentation

Maximum Marks: 5 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 ACOE	Rising Waters – ML-Based Flood Prediction System

1. Project Overview

Attribute	Details
Project Title	Rising Waters – Machine Learning Approach to Flood Prediction
Team ID	Batch 2023-27 ACOE
College	Anurag College of Engineering
Team Members	Venkatesh Balireddy (Team Lead) Archana Dhanani (Member) Shivatmika Gandikota (Member) Gode Siva Ramakrishna Durgaprasad (Member) Bavisetty Gopi Krishna (Member)
Technology Track	AI / Machine Learning
Submission Date	15 March 2026

2. Problem Statement

Floods are among the most devastating natural disasters globally, claiming thousands of lives and displacing millions every year. Despite their recurring nature, the lack of timely and accurate early-warning systems continues to amplify their destructive impact. Conventional forecasting methods often fall short in predicting floods at the right time, leaving authorities and communities with insufficient opportunity to respond. This project addresses that gap by building a machine learning-powered flood prediction system trained on historical weather data.

3. Dataset Details

Attribute	Details
Dataset Size	5,000 records
Input Features	annual_rainfall, seasonal_rainfall, cloud_visibility, humidity, temperature, river_level
Target Variable	flood_occurred (0 = No Flood, 1 = Flood)
Source	Synthetic dataset based on real Indian meteorological patterns
Split	80% Training (4,000 rows) 20% Testing (1,000 rows)

4. Machine Learning Pipeline

Step	Activity	Output
1	Environment Setup	Python 3.11, all dependencies installed
2	Dataset Generation	flood_dataset.csv — 5,000 rows, 6 features
3	Exploratory Data Analysis	4 visualisation charts (distribution, heatmap, box, pairplot)
4	Preprocessing	Imputed nulls, capped outliers, scaled with StandardScaler
5	4-Algorithm Training & Evaluation	DT 78.17%, KNN 80.67%, RF 82.00%, XGBoost 96.55%
6	Best Model Selection & Serialization	floods.save (XGBoost) + scaler.save (StandardScaler)
7	Flask App + IBM Cloud Deployment	Live 6-page web application on IBM Cloud

5. Web Application Pages

Page	Route	Description
Home Dashboard	/	Live prediction count, model accuracy comparison (DT/RF/KNN/XGB)
Prediction Form	/predict	6-field form: annual_rainfall, seasonal_rainfall, cloud_visibility, humidity, temperature, river_level
Flood Chance Result	/result/flood	Red alert banner, probability score, alert level, 6 emergency action cards
No Flood Result	/result/noflood	Green safe banner, all-clear badge, monitoring recommendations
Prediction History	/history	Full timestamped audit log from predictions_log.csv
About	/about	5-layer system architecture, model info, team details, use-case scenarios

6. System Architecture Summary

Layer	Technology
User Layer	Web Browser (Chrome, Firefox, Edge)
Presentation Layer	Jinja2 HTML Templates (6 pages)
Application Layer	Python 3.11 + Flask 2.x
ML Layer	XGBoost 1.7 + Scikit-learn 1.3 + Joblib
Data Layer	Pandas CSV (flood_dataset.csv + predictions_log.csv)
Deployment Layer	Docker + IBM Cloud Foundry

7. Team Contributions

S.No	Team Member	Role	Primary Contribution
1	Venkatesh Balireddy	Team Lead	Project lead, Flask app, IBM Cloud deployment
2	Archana Dhanani	Member	EDA, data visualisation, input validation utils
3	Shivatmika Gandikota	Member	Data preprocessing, StandardScaler pipeline
4	Gode Siva Ramakrishna Durgaprasad	Member	ML model training, XGBoost tuning, evaluation
5	Bavisetty Gopi Krishna	Member	HTML templates, CSS dark-theme UI design